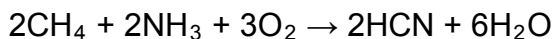


Answer **all** the questions in the spaces provided.

- 1 Hydrogen cyanide was first isolated from a blue dye (Prussian blue) which had been known since 1704 and is now known to be a coordination polymer, $\text{Fe}^{\text{III}}_4[\text{Fe}^{\text{II}}(\text{CN})_6]_3$.

In the year 2000, 732,552 tonnes of HCN were produced in the US. The most important process for HCN production is the Andrussov oxidation in which methane and ammonia react in the presence of oxygen at about 1200 °C:



The energy needed for the reaction is provided by the partial oxidation of methane and ammonia. (1 tonne = 1 000 kg)

HCN is also obtainable from fruits that have a pit, such as cherries, apricots, apples, and bitter almonds. Many of these pits contain small amounts of cyanohydrins such as mandelonitrile and amygdalin, which slowly release hydrogen cyanide. An article mentioned that 100 g of crushed apple seeds can yield about 10 mg of HCN.

- (a) (i) Draw the structure of the anion of Prussian blue to show its shape. Indicate on the structure the bond angle and give the name of the shape.

Structure:

Shape:

- (ii) Using specific examples from the information above, state **two** characteristic properties of transition elements.

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(iii) Find the percentage by mass of cyanide in Prussian blue.

[5]

(b) (i) Assuming that the amount of HCN produced in the US annually is the same, calculate the number of moles of HCN produced in the US from 2000-2002.

(ii) What is volume in dm^3 of oxygen gas required via the Andrussov oxidation to form the amount of HCN mentioned in **(i)** at r.t.p.?

[2]

- (c) An apple has an average of 4 seeds and each seed weighs an average of 0.20 g. Find out the mass in mg of HCN that can be yielded from seeds from 10 apples.

[1]

- (d) (i) HCN gas is a highly poisonous gas that can cause permanent damage to the central nervous system. The toxic level of HCN gas in the air is about 0.001 mg dm^{-3} .

Express the toxic level of HCN gas in the air in terms of mol dm^{-3} .

- (ii) How many molecules of HCN are present at this toxic level in 2 dm^3 ?

- (iii) When it rains, HCN dissolves in water to give an acid solution. If 2 dm^3 of air containing the toxic level of HCN dissolves in 20 dm^3 of rainwater, find the concentration of HCN in the acid solution.

[3]

- (e) Name and describe the mechanism for the formation of cyanohydrins starting from propanal and HCN. State any additional reagent that is needed.

Name:

Additional reagent:

Mechanism:

[3]

[Total: 14]

2 This question is about Group VII hydrides, HX.

- (a) The Group VII hydrides show clear trends in the H–X bond energy where X = Cl, Br and I. They behave as weak acids in liquid ethanoic acid, dissociating to different extents.

- (i) Write an equation to show how HX dissociates in ethanoic acid.

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- (ii) Rank the pK_a of the Group VII hydrides in ethanoic acid in decreasing order. Explain your answer with reference to the *Data Booklet*.

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- (iii) Suggest why the Group VII hydrides are weak acids in ethanoic acid whereas they are strong acids in water.

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[4]

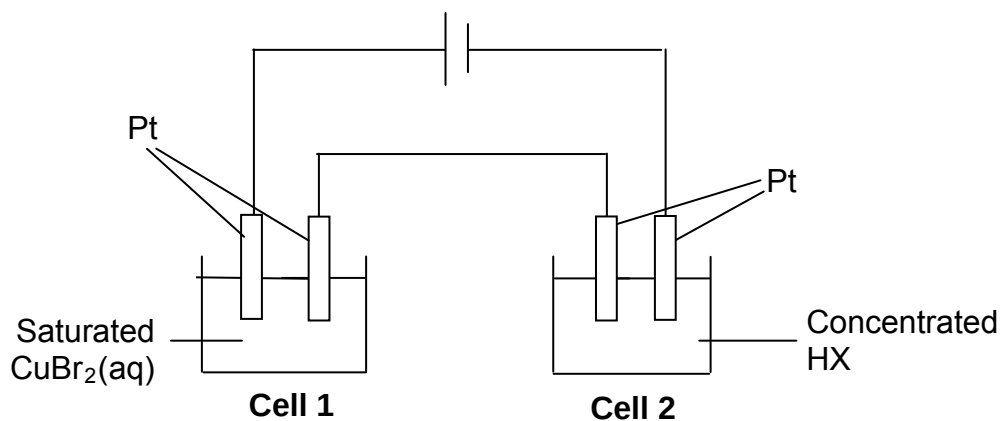
- (b) Another Group VII hydride, HF, behaves as a weak acid. A solution is prepared by dissolving HF in a fixed volume of water. The addition of some barium hydroxide to the solution neutralizes one third of the HF present. The pH of the resulting solution is 2.88.

- (i) Calculate the concentration of hydrogen ions in the solution after barium hydroxide is added.

- (ii) Using your answer to (i) and the given information, calculate the value of the acid dissociation constant, K_a , of HF.

[3]

- (c) The figure below shows two cells connected in series. **Cell 1** and **Cell 2** involve the electrolysis of saturated solution of CuBr_2 and concentrated HX respectively, where X is one of the Group VII elements.



- (i) Using $E^\ominus$ values, write ion-electron equations for the reactions occurring at the cathode and anode in **Cell 1** during electrolysis. Include state symbols.

Cathode:

Anode:

After a current was passed through the saturated solution of CuBr_2 in **Cell 1** for 10 minutes, the solution surrounding the anode was removed and it was found to require 35.50 cm^3 of $0.500 \text{ mol dm}^{-3}$ of sodium thiosulfate solution for complete reaction.

- (ii) Determine the number of moles of $\text{Na}_2\text{S}_2\text{O}_3$ used.

- (iii) Calculate the number of moles of electrons transferred.

- (iv) Hence, determine the current that flows through the circuit.

[5]

- (d) A halogen is formed at one of the electrodes in **Cell 2** after passing the current through it for 10 minutes and is collected by a syringe. It is found that the mass of the syringe increased by 5.04 g. Identify X by calculating the M_r of the halogen produced.

[2]

[Total: 14]

- 3 In nature, chlorine is abundantly found as the chloride ion, a component of the salt that is deposited in the earth or dissolved in the oceans.

The melting points of the chlorides of the elements sodium to phosphorus of the third period of the Periodic Table are given below.

compound	sodium chloride	magnesium chloride	aluminium chloride	silicon tetrachloride	phosphorus pentachloride
Melting point /°C	808	714	178*	-70	162
pH of solution	7	6.5	3	2	2

* sublimes at 178°C

- (a) (i) Account for the difference in the melting points of sodium chloride and silicon tetrachloride, in terms of structure and bonding.

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- (ii) Explain briefly the difference in the pH values when MgCl_2 , AlCl_3 and PCl_5 are separately dissolved in water. Write equations to support your answer.

[6]

- (b)** The reaction of silicon tetrachloride with water produces large amount of heat.

When 3 cm^3 of silicon tetrachloride is added to 100 cm^3 of water at 5.0°C , the temperature of the water rises to 9.2°C .

In the calculations below, you may take the density of SiCl_4 to be 1.483 g cm^{-3} and the specific heat capacity of the solution to be $4.2 \text{ J }^\circ\text{C}^{-1} \text{ cm}^{-3}$.

- (i)** Calculate the number of moles of silicon tetrachloride used.
- (ii)** Determine the heat evolved for the above reaction between silicon tetrachloride and water.
Assume the volume change is negligible on the addition of silicon tetrachloride.
- (iii)** Hence, calculate the enthalpy change of reaction, in kJ mol^{-1} .

[3]

(c) When solid aluminium chloride is heated above 180 °C, a vapour is formed which has $M_r = 267$. When this vapour is heated above 800 °C, the vapour has $M_r = 133.5$.

(i) Draw the displayed formula of the vapour at 180 °C, to show its bonding.

(ii) The average M_r of the vapour is found to be 240 at 600°C. Determine the molar composition of the vapour.

[2]

(d) Chlorine forms an oxide, Cl_2O_6 , which exists as singly charged ions in the solid state. The oxidation states of chlorine are +5 and +7 in the cation and anion respectively. Suggest the formulae of the ions and draw the dot-and-cross diagram for the **cation**.

Formulae of ions: and

Dot and cross diagram:

[2]

(e) When 100 cm^3 of chlorine gas measured at room temperature and pressure is bubbled into hot aqueous sodium hydroxide solution, a chlorine oxoanion is formed.

(i) Write the overall equation for the reaction that occurred.

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(ii) Calculate the maximum mass of the chlorine oxoanion formed.

[3]

[Total: 16]

4 (a) A solution of aminoethanoic acid at its isoelectric point can act as a buffer.

(i) By writing appropriate equations, show how a solution of aminoethanoic acid at its isoelectric point can function as a buffer.

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(ii) Predict the movement of aminoethanoic acid when a potential difference is applied at pH 13.

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[3]

(b) The table below shows the R groups of the various amino acids:

amino acid	aspartic acid	serine	cysteine	alanine	Lysine
Abbreviation	asp	ser	cys	ala	Lys
R group	$\text{CH}_2\text{CO}_2\text{H}^-$	$-\text{CH}_2\text{OH}$	$-\text{CH}_2\text{SH}$	$-\text{CH}_3$	$-(\text{CH}_2)_4\text{NH}_2^+$

A hexapeptide **A** has the following primary structure where asp has a free amine group:

asp-ser-cys-ala-lys-ala

(i) Draw the structural formula of hexapeptide **A** at pH 2.

(ii) Briefly describe the **two** types of R group interactions that would occur between the residues of hexapeptide **A** in an aqueous solution.

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(iii) Describe a simple chemical test to distinguish between **aqueous solutions** of alanine and serine.

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- (c) (i) Draw a labelled diagram of the β -pleated sheet, using R as the amino acid side chain. [6]

- (ii) Where, in a globular protein, will all the R groups of these amino acid residues of hexapeptide **A** be located?

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[4]

- (d) Write an equation to show the enzymic hydrolysis of a peptide bond.

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[1]

- (e) A particular proteolytic enzyme **X** can hydrolyse the peptide bond between carboxyl group of cysteine and amino group of alanine of a pentapeptide to give the amino acid, cys, and two similar dipeptides, ala-cys.

Another proteolytic enzyme **Y** can hydrolyse the peptide bond between carboxyl group of alanine and amino group of cysteine to give the amino acid, cys, and two similar dipeptides, cys-ala.

Deduce the primary structure of this pentapeptide.

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[2]

[Total: 16]